

711_Predictive - Problem Set -2

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1. Problem to demonstrate that the population regression line is fixed, but least square regression line varies.

Suppose the population regression line is given by $Y = 2 + 3x$, while the data comes from the model $y = 2 + 3x + \epsilon$.

Step 1: For x in the range $[5,10]$ graph the population regression line.

Step 2: Generate $x_i (i = 1, 2, \dots, n)$ from Uniform $(5, 10)$ and $\epsilon_i (i = 1, 2, \dots, n)$ from $N(0, 4^2)$. Hence, compute $y_1, y_2, \dots, y_n$.

Step 3: On the basis of the data $(x_i, y_i) (i = 1, 2, \dots, n)$ generated in **Step 2**, report the least squares regression line.

Step 4: Repeat steps 2-3 five times. Graph the 5 least squares regression lines over the population regression line obtained in Step 1. Interpret the findings. Take $n = 50$. Set the seed as seed=123.

Solution 1:

The population regression function is defined as:

$$Y = 2 + 3x$$

The sample regression function including the random error can be defined as:

$$y_i = 2 + 3x_i + \epsilon_i$$

Where:

- $x_i \sim \text{Uniform}(5,10)$
- $\epsilon_i \sim N(0, 4^2)$

```
rm(list=ls())
set.seed(123)

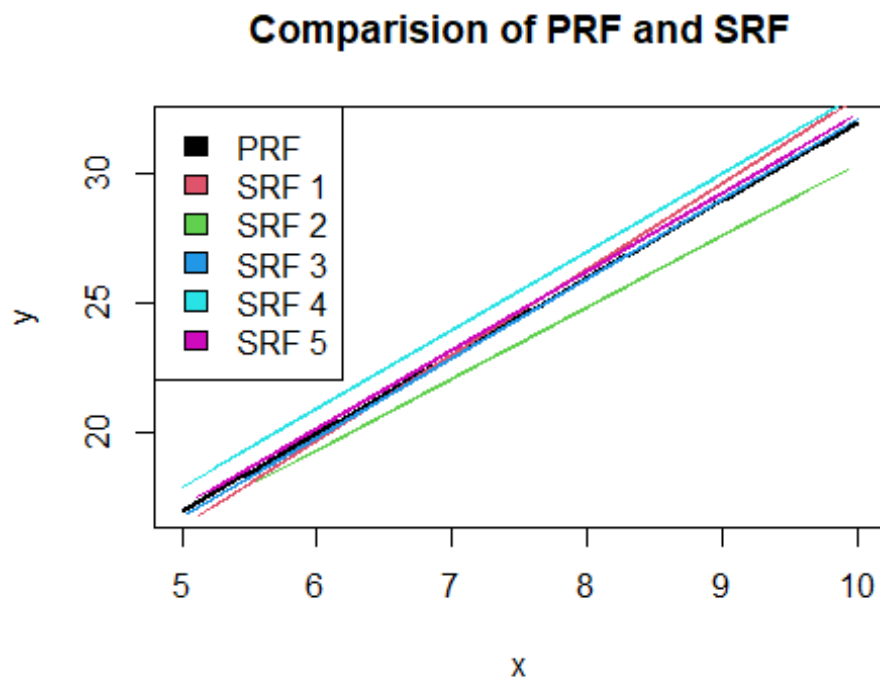
x = seq(5,10,length.out= 200)
y = 2+3*x
plot(x,y,col = "black", lwd = 2,
     main = "Comparison of PRF and SRF",type="l")
beta_hat = matrix(0,1,2)
for(i in 1:5)
{
  x = runif(50,5,10)
  e = rnorm(50,0,4)
```

```

y_sam = 2+ 3*x+ e

model = lm(y_sam~x)
summary(model)
beta_hat = rbind(beta_hat,model$coefficients)
lines(x,predict(model), col= i+1,lwd=1)
}
legend("topleft",legend= c("PRF",paste("SRF",1:5)),fill =
c("black",2:6),border = "black")

```



```

print("The estimates in 5 cases are given as:")
## [1] "The estimates in 5 cases are given as:"
beta_hat[-1,]

##      (Intercept)      x
## [1,] -0.09638929  3.305396
## [2,]  2.79218839  2.761042
## [3,]  1.39299737  3.073267
## [4,]  2.82308856  3.023608
## [5,]  2.03250638  3.028097

```

Interpretation

Although the population regression line is fixed, the least squares regression line varies from sample to sample due to random error (sampling fluctuations).

2. Problem to demonstrate that β_0 and β minimise RSS

Step 1: Generate x_i from $Uniform(5,10)$ and mean centre the values. Generate ϵ_i from $N(0,1)$. Calculate $y_i = 2 + 3x_i + \epsilon_i$, $i = 1, 2, \dots, n$. Take $n=50$ and $seed=123$.

Step 2: Now imagine that you only have the data on (x_i, y_i) , $i = 1, 2, \dots, n$, without knowing the mechanism that was used to generate the data in step 1.

Assuming a linear regression of the type $y_i = \beta_0 + \beta x_i + \epsilon_i$, and based on these data (x_i, y_i) , $i = 1, 2, \dots, n$. Obtain the least squares estimates of β_0 and β .

Step 3: Take a large number of grid values of (β_0, β) that also include the least squares estimates obtained from step 2. Compute the RSS for each parametric choice of (β_0, β) .

where:

$$RSS = (y_1 - \beta_0 - \beta x_1)^2 + (y_2 - \beta_0 - \beta x_2)^2 + \dots + (y_n - \beta_0 - \beta x_n)^2.$$

Find out for which combination of (β_0, β) , RSS is minimum.

Solution 2:

```
rm(list=ls())
set.seed(123)

x = runif(50,5,10)
x_mc = x - mean(x)

e = rnorm(50,0,1)
y_sam = 2 + 3*x_mc + e

b_0 = seq(1,3, 0.01)
b_1 = seq(2,4, 0.01)

b = expand.grid(b_0,b_1)
head(b)

##   Var1 Var2
## 1 1.00    2
## 2 1.01    2
## 3 1.02    2
## 4 1.03    2
## 5 1.04    2
## 6 1.05    2

rss = numeric(length(b[,1])) # Pre-allocate vector for speed

for(i in 1:length(b[,1])) {
  # RSS formula: sum of squared residuals
  rss[i] = sum((y_sam - (b[i,1] + b[i,2] * x_mc))^2)
```

```

}
output = data.frame(b,rss)
colnames(output) = c("B_0", "B_1", "RSS")

print("Minimum RRS and the values of the estimates")

## [1] "Minimum RRS and the values of the estimates"

output[which.min(output$RSS), ]

##           B_0  B_1      RSS
## 21815  2.06  3.08 42.44764

coefs = lm(y_sam ~ x_mc)$coefficients
rss_min = sum((y_sam - (coefs[1] + coefs[2]*x_mc))^2)
rss_min

## [1] 42.4455

```

We see that the RSS is minimum at the least squares estimates of β_0 and β .

3. Problem to demonstrate that least square estimators are unbiased

Step 1: Generate $x_i (i = 1, 2, \dots, n)$ from $Uniform(0,1)$, $\epsilon_i (i = 1, 2, \dots, n)$ from $N(0,1)$ and hence generate y using $y_i = \beta_0 + \beta x_i + \epsilon_i$.

(Take: $\beta_0 = 2, \beta = 3$).

Step 2: On the basis of the data $(x_i, y_i) (i = 1, 2, \dots, n)$ generated in Step 1, obtain the least square estimates of $\widehat{\beta}_0$ and $\widehat{\beta}$. Repeat Steps 1-2, $R = 1000$ times. In each simulation obtain $\widehat{\beta}_0$ and $\widehat{\beta}$.

Finally, the least-square estimates will be given by the average of these estimated values. Compare these with the true β_0 and β and comment. Take $n = 50$ and seed = 123.

Solution 3

```

rm(list=ls())
set.seed(123)
beta_hat = matrix(0,1,2)
for(i in 1:1000)
{
  x = runif(50,5,10)
  e = rnorm(50,0,4)
  y_sam = 2+ 3*x+ e

  model = lm(y_sam~x)
  summary(model)
  beta_hat = rbind(beta_hat,model$coefficients)
}
apply(beta_hat[-1,],2,mean)

```

```
## (Intercept)          x
##      2.123765      2.985690
```

Comment

The average of the estimated coefficients, i.e. empirical $E(\widehat{\beta}_0) = 2.055461$ and $E(\widehat{\beta}) = 3$ is close to the true parameter values $\beta_0 = 2$ and $\beta = 2.993070$, demonstrating that in the long run, $\widehat{\beta}_0$ and $\widehat{\beta}$ are unbiased for β_0 and β .

4. Comparing several simple linear regressions

Attach “Boston” data from MASS library in R. Select median value of owner occupied homes, as the response and per capita crime rate, nitrogen oxides concentration, proportion of blacks and percentage of lower status of the population as predictors:

- Selecting the predictors one by one, run four separate linear regressions to the data. Present the output in a single table.
- Which model gives the best fit?
- Compare the coefficients of the predictors from each model and comment on the usefulness of the predictors.

Solution 4

Response Variable (Y):

- medv: Median value of owner-occupied homes.

Predictor Variables (X):

- crim: Per capita crime rate.
- nox: Nitrogen oxides concentration.
- black: Proportion of black people.
- lstat: Percentage of lower status of the population.

```
library(MASS)
data = Boston
attach(data)
library(stargazer)

##
## Please cite as:

## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary
## Statistics Tables.

## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

sub_data = data.frame(medv, crim, nox, black, lstat)
head(sub_data)
```

```
##   medv   crim   nox   black   lstat
## 1 24.0 0.00632 0.538 396.90  4.98
## 2 21.6 0.02731 0.469 396.90  9.14
## 3 34.7 0.02729 0.469 392.83  4.03
## 4 33.4 0.03237 0.458 394.63  2.94
## 5 36.2 0.06905 0.458 396.90  5.33
## 6 28.7 0.02985 0.458 394.12  5.21

model1 = lm(medv~crim)
model2 = lm(medv~nox)
model3 = lm(medv~black)
model4 = lm(medv~lstat)

stargazer(model1,model2,model3,model4,type="text")

##
## =====
##                               Dependent variable:
##                               -----
##                               medv
##                               (1)      (2)      (3)      (4)
## -----
## crim                        -0.415***
##                               (0.044)
##
## nox                          -33.916***
##                               (3.196)
##
## black                        0.034***
##                               (0.004)
##
## lstat                        -0.950***
##                               (0.039)
##
## Constant                    24.033*** 41.346*** 10.551*** 34.554***
##                               (0.409)  (1.811)  (1.557)  (0.563)
##
## -----
## Observations                 506      506      506      506
## R2                           0.151      0.183      0.111      0.544
## Adjusted R2                  0.149      0.181      0.109      0.543
## Residual Std. Error (df = 504) 8.484      8.323      8.679      6.216
## F Statistic (df = 1; 504)      89.486*** 112.591*** 63.054*** 601.618***
## =====
## Note:                        *p<0.1; **p<0.05; ***p<0.01
```

Using the stargazer function, we obtain a table containing the simple linear regression coefficient and other summary measures.

Comments

- Here model4, where medv is explained by lstat, has the highest R^2 value, 0.544 .
- The coefficients indicate the direction and strength of the relationship between the response and each predictor. Among these, {lstat} shows the strongest association with {medv}.
- With the stargazer package we can see the number of stars on the estimates and comment that each of simple linear equations are significant in the t-tests at 0.01 level of significance.